Psych-Bench: Supplementary Material

Complete Benchmark Inventory

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Table S1: Complete Benchmark Inventory

Table S1 lists all 84 benchmarks in the Psych-Bench suite, organized by domain. Each entry provides the benchmark name, source paradigm, primary citation, year, key metric, and HDC stimulus adapter type used for encoding.

Table 1: Complete inventory of all 84 benchmarks in Psych-Bench, organized by domain.

Domain	**Benchmark**	**Paradigm / Citation**	**Year**	**Key Metric**	**HDC Adapter**
Executive	Stroop	Color-word interference (Stroop, 1935)	1935	stroop_effect	Sequence
Executive	WCST	Set-shifting (Berg, 1948)	1948	categories_completed	Sequence
Executive	Flanker	Response conflict (Eriksen and Eriksen, 1974)	1974	flanker_effect	Sequence
Executive	Iowa Gambling	Decision under ambiguity (Bechara et al., 1994)	1994	overall_net_score	Reward
Executive	Ravens	Fluid intelligence (Raven, 1938)	1938	overall_accuracy	Spatial
Executive	Tower of London	Planning (Shallice, 1982)	1982	overall_optimal_rate	Spatial
Executive	Dual-Task	Divided attention (Pashler, 1994)	1994	dual_task_cost	Sequence
WorM	N-back	Working memory updating (Jaeggi et al., 2010)	1958	nback_2::accuracy	Sequence
WorM	Change Detection	Visual WM capacity (Luck and Vogel, 1997)	1997	change_detection_k4	Spatial

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Domain	Benchmark	Paradigm / Citation	Year	Key Metric	HDC Adapter
WorM	Serial Recall	Short-term memory (Murdock, 1962)	1962	primacy_advantage	Sequence
WorM	Spatial Updating	Spatial WM (Oberauer, 2003)	2003	updating_accuracy	Spatial
WorM	Binding	Feature binding (Luck and Vogel, 1997)	1997	binding_accuracy	Spatial
WorM	Digit Span	Verbal WM (Wechsler, 2008)	1997	forward_span	Sequence
MemoryAgent	Accurate Retrieval	Retrieval accuracy (Anderson, 1983)	1983	retrieval_accuracy	Scenario
MemoryAgent	Test-Time Learning	Test-enhanced learning	2006	online_gain	Scenario
MemoryAgent	Long-Range	Long-range retention	1885	retention_accuracy	Scenario
MemoryAgent	Conflict Resolution	Interference resolution	1996	resolution_accuracy	Scenario
MemoryAgent	Prospective Memory	Future intention execution	1990	pm_hit_rate	Scenario
Attention	Visual Search	Feature/conjunction search	1980	feature_search_accuracy	Spatial
Attention	Attentional Blink	Dual-target RSVP	1992	t1_accuracy	Sequence
SustAttn	SART	Sustained attention (Robertson et al., 1997)	1997	sart_d_prime	Sequence
SustAttn	PVT	Vigilance	1985	pvt_mean_rt	Sequence
SustAttn	CPT	Continuous performance	1956	cpt_hit_rate	Sequence
Social	RME	Emotion recognition (Baron-Cohen et al., 2001)	2001	rme_accuracy	Scenario
Social	Ultimatum Game	Social fairness (Güth et al., 1982)	1982	offer_ratio	Reward
Social	Social Norm	Norm compliance	2006	norm_detection_accuracy	Scenario
ToMBench	False Belief	False belief task (Baron-Cohen et al., 1985)	1983	false_belief_accuracy	Scenario
ToMBench	Faux Pas	Faux pas detection (Stone et al., 1998)	1999	faux_pas_accuracy	Scenario
ToMBench	Persuasion	Persuasion evaluation	1994	persuasion_detection	Scenario
ToMBench	Strange Story	Strange stories (Happé, 1994)	1994	strange_story_accuracy	Scenario
ToMBench	Hinting	Hinting task (Corcoran et al., 1995)	1995	hinting_accuracy	Scenario

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Domain	Benchmark	Paradigm / Citation	Year	Key Metric	HDC Adapter
Motor	SRTT	Implicit sequence learning (Nissen and Bullemer, 1987)	1987	sequence_rt_ticks	Sequence
Motor	Fitts' Law	Speed-accuracy tradeoff (Fitts, 1954)	1954	fitts_r_squared	Spatial
Motor	Bimanual	Bimanual coordination	1984	coordination_cost	Spatial
Language	Garden-Path	Sentence processing (Ferreira et al., 2002)	1982	garden_path_accuracy	Semantic
Language	Semantic Coherence	Text comprehension	2004	coherence_mean	Semantic
Language	Lexical Decision	Word recognition	1971	lexicality_effect	Semantic
Language	Semantic Priming	Priming facilitation	1977	priming_effect	Semantic
Metacognition	Calibration	Confidence calibration	1982	calibration_error_ece	Sequence
Metacognition	Feeling of Knowing	FOK judgment	1965	fok_gamma	Sequence
Reasoning	ARC Fluid	Abstract reasoning (Chollet, 2019)	2019	transfer_accuracy	Spatial
Reasoning	ARC Compositional	Compositional rules	2021	compositional_accuracy	Spatial
Reasoning	ARC Analogy	Analogical transfer	2019	analogy_accuracy	Spatial
Reasoning	ARC Abductive	Abductive inference	2019	abductive_accuracy	Spatial
Reasoning	ARC Algebra	Algebraic encoding	2003	algebra_accuracy	Spatial
Reasoning	ARC Chain	Multi-step chaining	2019	chain_accuracy	Spatial
Reasoning	ARC FewShot	Few-shot learning	2019	few_shot_accuracy	Spatial
Reasoning	ARC Noise	Noise robustness	2019	noise_accuracy	Spatial
Reasoning	ARC RSA	Representational similarity	2008	rsa_r_squared	Spatial
Reasoning	ARC Scaling	Parametric scaling	2009	capacity_threshold	Spatial
Reasoning	ARC Staircase	Adaptive threshold	1971	staircase_threshold	Spatial
CogBench	Probabilistic Reasoning	Beads task	1966	draws_to_decision	Sequence
CogBench	Horizon Task	Explore-exploit	2014	directed_exploration	Reward
CogBench	Restless Bandit	Multi-armed bandit	2006	total_reward	Reward
CogBench	Instrumental Learning	Associative learning	2011	learning_rate	Reward

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Domain	Benchmark	Paradigm / Citation	Year	Key Metric	HDC Adapter
CogBench	Two-Step	MB/MF learning (Daw et al., 2011)	2011	model_basedness	Reward
CogBench	Temporal Discounting	Delay discounting	2004	discount_rate	Reward
CogBench	BART	Risk-taking (Lejuez et al., 2002)	2002	avg_pumps	Reward
CogBench	Reversal Learning	Flexible learning (Cools et al., 2002)	2002	win_stay_rate	Reward
Affect	Valence Classification	Affective valence	1980	valence_accuracy	Scenario
Affect	Mood-Congruent Recall	Mood-memory interaction (Bower, 1981)	1981	congruence_ratio	Scenario
Affect	Emotional Stroop	Affective interference (Williams et al., 1996)	1996	emotional_interference	Sequence
Creativity	Remote Associates	RAT (Mednick, 1962)	1962	solution_rate	Scenario
Creativity	Alternate Uses	Divergent thinking	1967	fluency_score	Scenario
Inhibition	Go/No-Go	Response inhibition	1969	go_accuracy	Sequence
Inhibition	Stop-Signal	Response stopping (Logan and Cowan, 1984)	1984	sst_stop_accuracy	Sequence
Butlin	Consciousness Indicators	14-indicator battery (Butlin et al., 2023)	2023	indicators_present	—
Neuromod	Reward Learning	DA prediction error (Schultz et al., 1997)	1997	rpe_correlation	Reward
Neuromod	Yerkes-Dodson	NE inverted-U (Yerkes and Dodson, 1908)	1908	inverted_u_fit_r2	—
Neuromod	Attention Network	ACh ANT decomposition	1990	conflict_effect	Sequence
Neuromod	Mood Induction	5-HT mood-cognition	2009	mood_perf_r	Scenario
Neuromod	Pharm. Ablation	Transmitter knockout	2002	knockout_d	—
Neuromod	Pharm. Challenge	Agonist/antagonist	2011	inverted_u_fit	—
Neuromod	Injection Challenge	PK decay + tolerance	2011	peak_shift	—
Neuromod	Allostatic Stress	Chronic load	1998	allostatic_index	—
Neuromod	Dose Response	Graded dose sweep	1937	monotonicity_score	—
Neuromod	Tolerance/Withdrawal	Sustained exposure	2001	tolerance_ratio	—

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Domain	**Benchmark**	**Paradigm / Citation**	**Year**	**Key Metric**	**HDC Adapter**
Neuromod	Antagonist Profiles	Named antagonists	2013	selectivity_index	—
Neuromod	Multi-Transmitter Synergy	Paired interactions	2002	synergy_coefficient	—
Neuromod	Consciousness Pharm.	Drug $\to$ Φ proxy	2004	proxy_peak	—
Neuromod	Consciousness Feedback	Φ round-trip	2014	feedback_latency	—
Neuromod	Moral Oxytocin	OT $\times$ moral topology	2012	moral_ot_r	—
Neuromod	Behavioral Knockout	Knockout effect size	2002	behavioral_d	—
Neuromod	Live-Loop Ablation	In-loop knockout	2002	behavioral_delta	—

Supplementary Data Files

The following CSV data files are included in the repository under `papers/data/psych_bench/` and are regenerated deterministically by:

```
cargo run -p symthaea-psych-bench -example psych_bench_paper_data
```

- `normative_zscores.csv` — Per-benchmark normative z-scores comparing agent performance against published human baselines (Figure 5 forest plot). Columns: benchmark, metric, agent value, human mean, human SD, z-score (sign-corrected), percentile, clinical band.
- `cognitive_profile.csv` — Domain-level cognitive profile scores (Figure 4 radar chart). Eight domains with score, benchmark count, and qualitative interpretation.
- `reliability.csv` — Test-retest reliability (ICC) for each benchmark across 3 sessions $\times$ 10 seed families (Table 2).
- `correlations.csv` — Cross-domain correlation matrix for 10 key mechanism pairs (Table 3).
- `ablation_domains.csv` — Domain scores under 6 ablation presets (Figure 7 grouped bar chart).
- `sat_*.csv` — Speed-accuracy tradeoff curves for individual benchmarks (Figure 8).
- `neuromod_profiles.csv` — Neuromodulator benchmark results including behavioral knockout effect sizes, dose-response monotonicity, tolerance/withdrawal dynamics, and consciousness pharmacology.

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